

```

69.         numNeighbors += 1 # Bottom-left neighbor is alive.
70.         if cells[(x, below)] == ALIVE:
71.             numNeighbors += 1 # Bottom neighbor is alive.
72.         if cells[(right, below)] == ALIVE:
73.             numNeighbors += 1 # Bottom-right neighbor is alive.
74.
75.         # Set cell based on Conway's Game of Life rules:
76.         if cells[(x, y)] == ALIVE and (numNeighbors == 2
77.            or numNeighbors == 3):
78.             # Living cells with 2 or 3 neighbors stay alive:
79.             nextCells[(x, y)] = ALIVE
80.         elif cells[(x, y)] == DEAD and numNeighbors == 3:
81.             # Dead cells with 3 neighbors become alive:
82.             nextCells[(x, y)] = ALIVE
83.         else:
84.             # Everything else dies or stays dead:
85.             nextCells[(x, y)] = DEAD
86.
87.     try:
88.         time.sleep(1) # Add a 1 second pause to reduce flickering.
89.     except KeyboardInterrupt:
90.         print("Conway's Game of Life")
91.         print('By Al Sweigart al@inventwithpython.com')
92.         sys.exit() # When Ctrl-C is pressed, end the program.

```

After entering the source code and running it a few times, try making experimental changes to it. The comments marked with (!) have suggestions for small changes you can make. On your own, you can also try to figure out how to do the following:

- Adjust the percentage of cells that start as living, instead of always using 50 percent.
- Add the ability to read in the initial state from a text file, so the user can edit the starting cell states manually.

Exploring the Program

Try to find the answers to the following questions. Experiment with some modifications to the code and rerun the program to see what effect the changes have.

1. What happens when you change `WIDTH = 79` on line 10 to `WIDTH = 7`?
2. What happens if you delete or comment out `print('\n' * 50)` on line 36?
3. What happens if you change `random.randint(0, 1)` on line 28 to `random.randint(0, 10)`?
4. What happens if you change `nextCells[(x, y)] = DEAD` on line 85 to `nextCells[(x, y)] = ALIVE`?